4. Solution Architecture

The Solution Architecture explains how the Electricity Consumption in Indian States Analysis Dashboard is designed. It shows the flow of data from the source to the final visualization dashboard used by users for analysis and decision making.

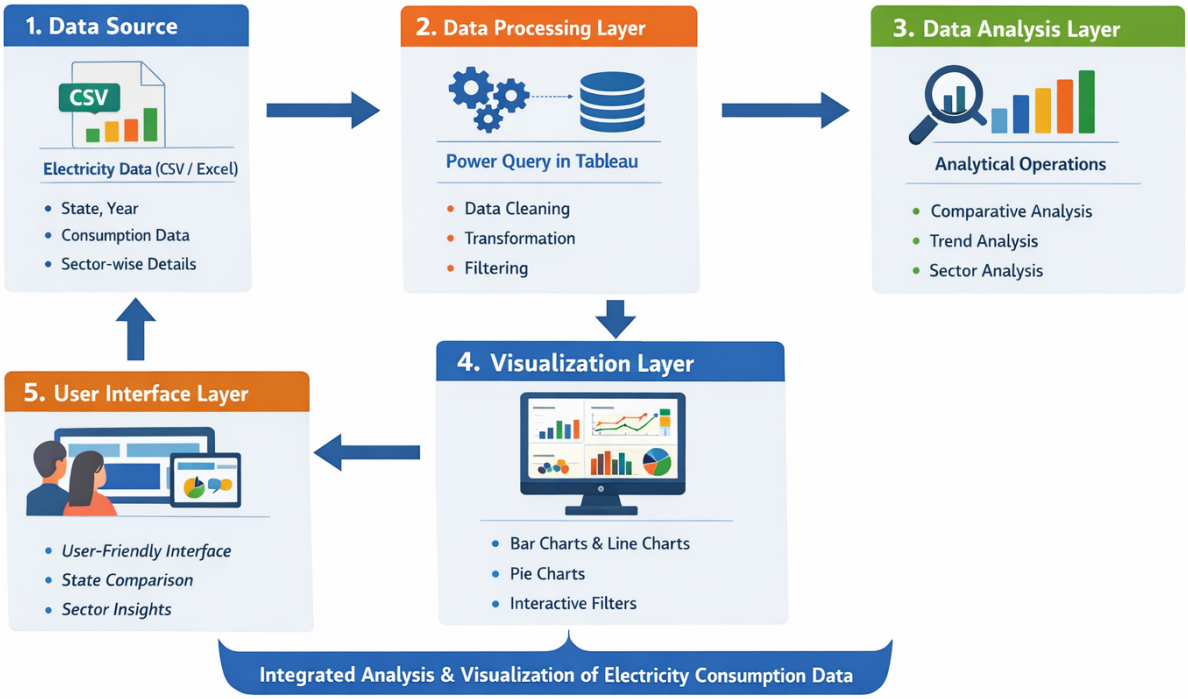
1. Data Source: The system uses electricity consumption datasets stored in CSV or Excel format. These datasets include information such as state name, year, electricity consumption values, and sector-wise consumption details like residential, industrial, commercial, and agricultural usage.

2. Data Processing Layer: In this stage, the raw data is cleaned and transformed using data preparation tools. Tasks such as removing duplicates, handling missing values, and converting data types are performed to make the dataset suitable for analysis.

3. Data Analysis Layer: After processing, analytical operations are applied to identify patterns and trends. This includes comparing electricity consumption across states, analyzing yearly trends, and studying sector-wise electricity usage.

4. Visualization Layer: The processed data is visualized using a dashboard with charts such as bar charts, line charts, and pie charts. Interactive filters allow users to explore different states, years, and sectors easily.

5. User Interface Layer: The final dashboard provides a user-friendly interface where users can view insights, compare electricity consumption among states, and understand overall electricity usage patterns.

4. Solution Architecture

Electricity Consumption in Indian States Analysis Dashboard